Improve One Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected repair, regression and candidate record.

Document the equality edit, operate the full set, diagnose a regression and audit the independent equipment revision.

Keep one habit: Keep defects, expected results, actual runs, version history and final-evaluation boundaries together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Test Edge Cases and Fairness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Test Edge Cases and Fairness

Build a Paper Prototype

Find a Real Problem

W1 · Record the exact equality repair

> Preserve the old failed trace and name every unchanged condition.

Your evidence record

- Standalone repair reference: catalogue B1=10 min quiet Yes, B2=8 No, B3=12 Yes, B4=15 No, in that order. Valid whole minutes 0–30, Quiet required Yes/No. Require duration≤minutes AND (quiet not required OR activity quiet), choose first fit, no match after all fail, clarify invalid input before search and reset each run. S0 uses < for time; S1 changes only that to ≤; T1 uses ≤ but omits quiet. R cases are (11,Yes),(9,No),(6,No),(9,Yes). Add equality (10,Yes), (8,No) and validation/endpoints (0,Yes),(30,Yes),(31,Yes),(blank,Yes),(10.5,Yes),(10,Maybe). B version 2 changes only B1 duration to 20; test (12,Yes). B version 3 uses durations 20,18,17,15 with original quiet labels/order; test (15,No) and (15,Yes). Label actual catalogue and prototype versions on every trace.

- Use S0's failure on 10 Yes with catalogue B version 1. Record the expected and incorrect **output**s, the faulty time comparison, the exact S1 correction and the other conditions that must remain.

  Explain what old **evidence** to preserve.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Operate repair and regression cases

> Use the supplied complete test reference and record actual runs separately from predictions.

Your evidence record

- Use the complete activity repair/test reference in W1 for every required request and catalogue variation. Keep the old S0 record, new S1 instructions, actual baseline/prototype runs and unrun cases separately labelled.

- Plan and operate the S1 **repair** and **regression** set described in the lesson, using the appropriate catalogue version for each case. Record expected and observed results separately with reasons and traces.

  Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures. Do not replace observed mistakes with worked answers.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Diagnose the introduced quiet failure

> Use the earlier passing case to establish regression.

Your evidence record

- T1 accepts equal times but omits the quiet check. Explain why it passes 10 Yes yet fails 9 Yes. Use S0's earlier correct result for 9 Yes to justify the word **regression**.

  What should change in T1, and which two cases should be rerun first?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Protect the meaning of unseen evaluation

> Retain old evaluation results and distinguish development after a revision.

Your evidence record

- After seeing a reserved final case fail, a team edits its candidate, reruns that same case and calls the result “a successful unseen test of the new version”. Explain what is wrong and what should be recorded instead.

  State what must be saved before an evaluation begins.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit a different proposed repair

> Keep one first attempt before hints or feedback.

Read and explain independently

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a different proposed repair

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

If the sorter has an **evidence**d defect, save a new version with one **evidence**-linked change and a before/after test table; otherwise retain the current tested version and state that no repair was needed.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check the **repair** and earlier behaviour for Improving and checking **regression**s.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Improving and checking **regression**s. Proposed explanation: “One **repair**ed example proves improvement everywhere”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your **repair** if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

The team changes lighting, labels and rules together; errors fall from four to one.

Can it identify the cause of improvement?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Dim photos fail. You change the lighting and all decision rules together; performance improves. Can you tell which change helped? Design a clearer comparison.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 35 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes.

U0 accepts only duration<available time. U1 fixes ≤ but accidentally removes the quiet requirement.

AI ACADEMY / AI-1/35

Improve One Cause

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Boundary failure

At 8 minutes, U0 rejects A and rejects noisy B: none.



2. Expected repair

Correct logic must allow A when time equals 8.



3. Regression case

At 6 quiet-required minutes, correct output is none.



4. Bad revision

U1 selects noisy B at 6; fixing equality introduced a new failure.

CHANGE ONE THING

Use U2, which changes < to <= and keeps the quiet check.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.
- Which labelled step matches this detail? U1 selects noisy B at 6; fixing equality introduced a new failure.
- 1. Bad revision
- 2. **Regression** case
- 3. Expected **repair**
- 4. Boundary failure
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bad revision U1 selects noisy B at 6; fixing equality introduced a new failure. A **repair** must preserve the full contract, not just make one failing example pass.

The quiet condition and duration comparison are separate requirements. Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior.

Save previously passing cases to reveal such **regression**s.

Changed-case question

- Changed-case practice: Use U2, which changes < to ≤ and keeps the quiet check. Which conclusion is supported by the supplied facts?
- 1. At 8 return A; at 6 return none.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: At 8 return A; at 6 return none. Equality is **repair**ed while B remains ineligible because it is noisy.

Repair and save your work

Review Test Edge Cases and Fairness (ai-1/34). Goal: Choose tests just below, at and above a decision boundary. Short practice: Which single boundary run most directly detects the strict-comparison bug, and why?.

Return to this week and explain what you changed.

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the **repair**?.

Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

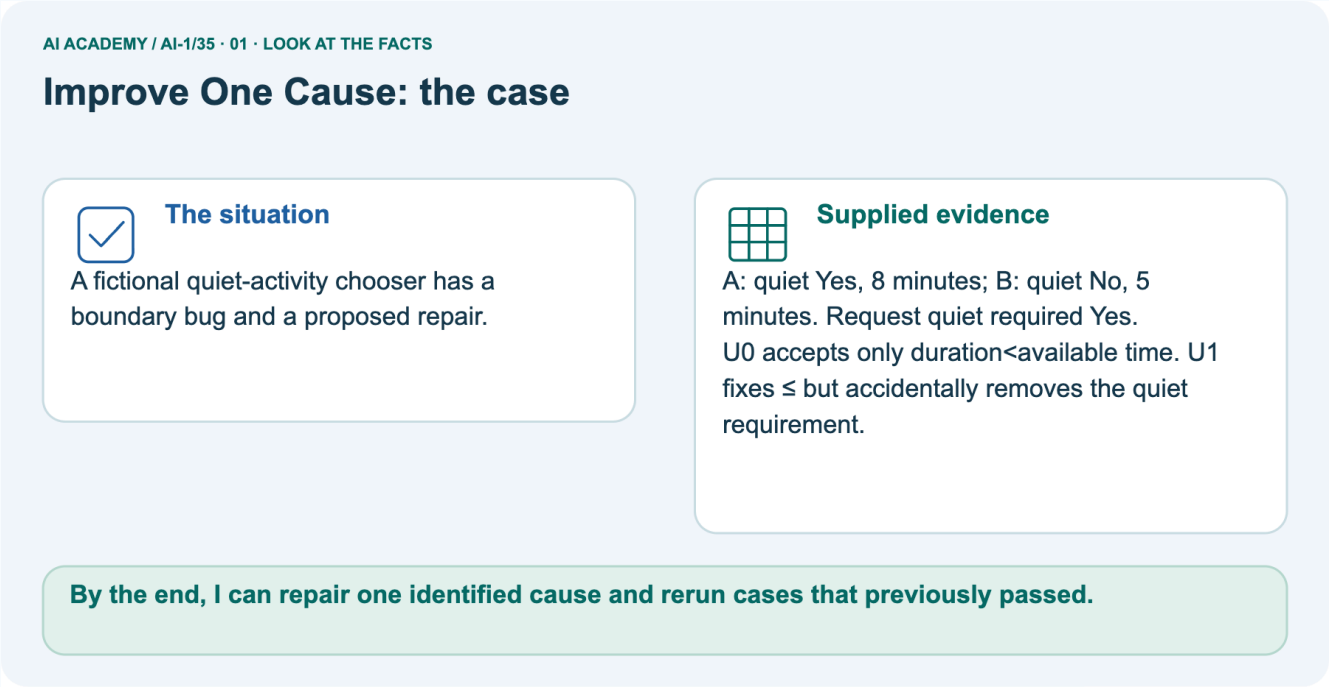
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Improve One Cause: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

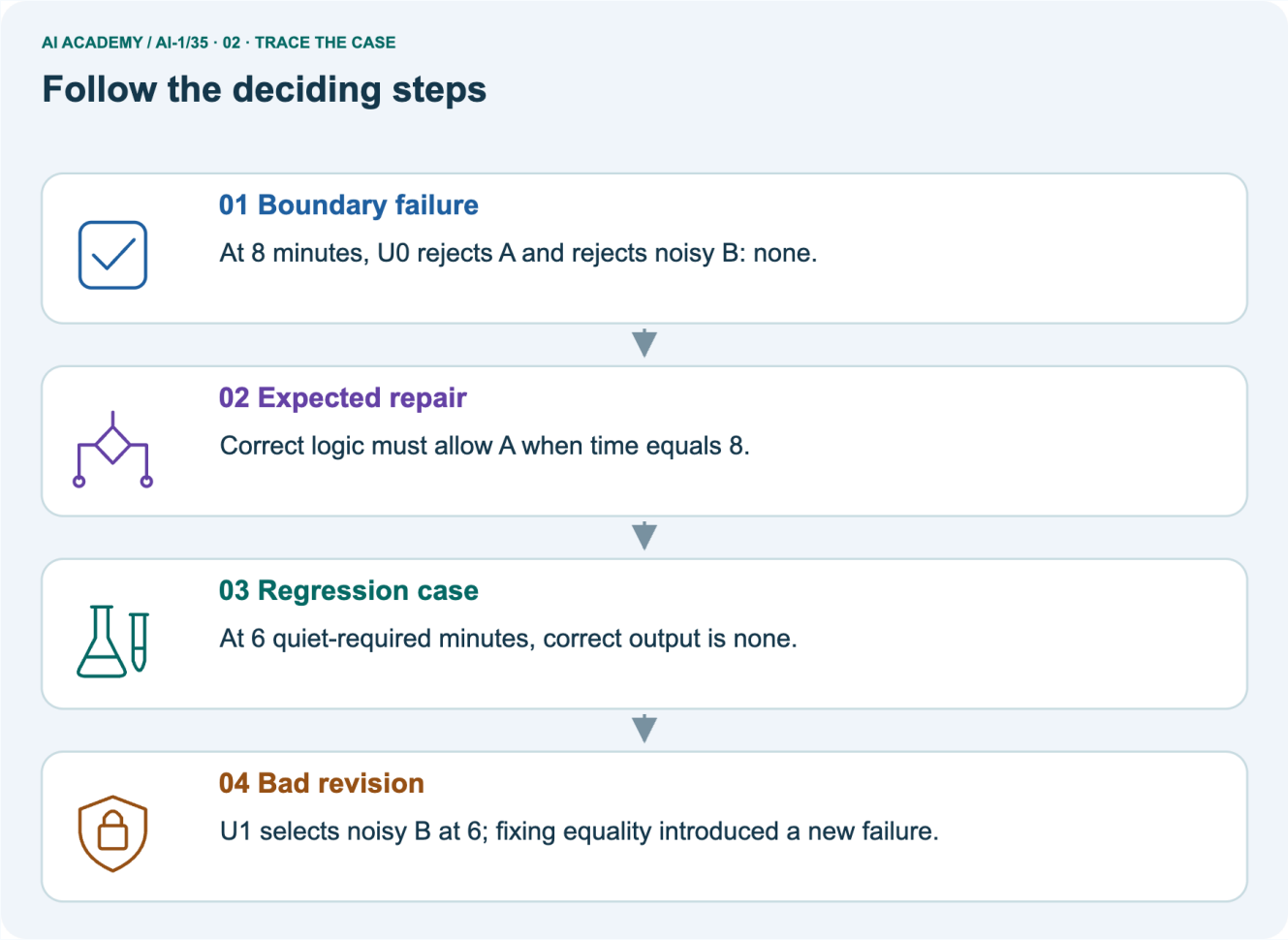
A fictional quiet-activity chooser has a boundary bug and a proposed **repair**. Supplied facts: A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes. U0 accepts only duration<available time.

U1 fixes ≤ but accidentally removes the quiet requirement.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



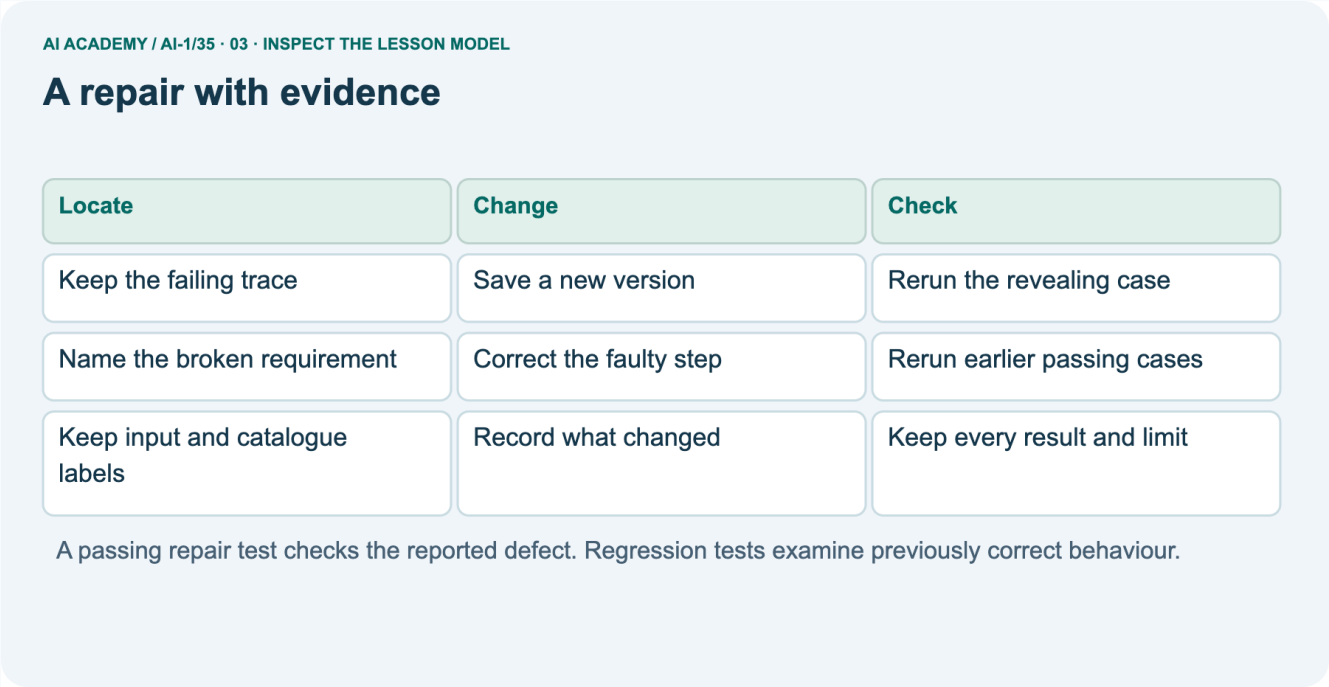
Follow the labelled picture with the matching explanation. Open the vector diagram

Boundary failure: At 8 minutes, U0 rejects A and rejects noisy B: none. → Expected **repair**: Correct logic must allow A when time equals 8. → **Regression** case: At 6 quiet-required minutes, correct **output** is none. → Bad revision: U1 selects noisy B at 6; fixing equality introduced a new failure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A repair with evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

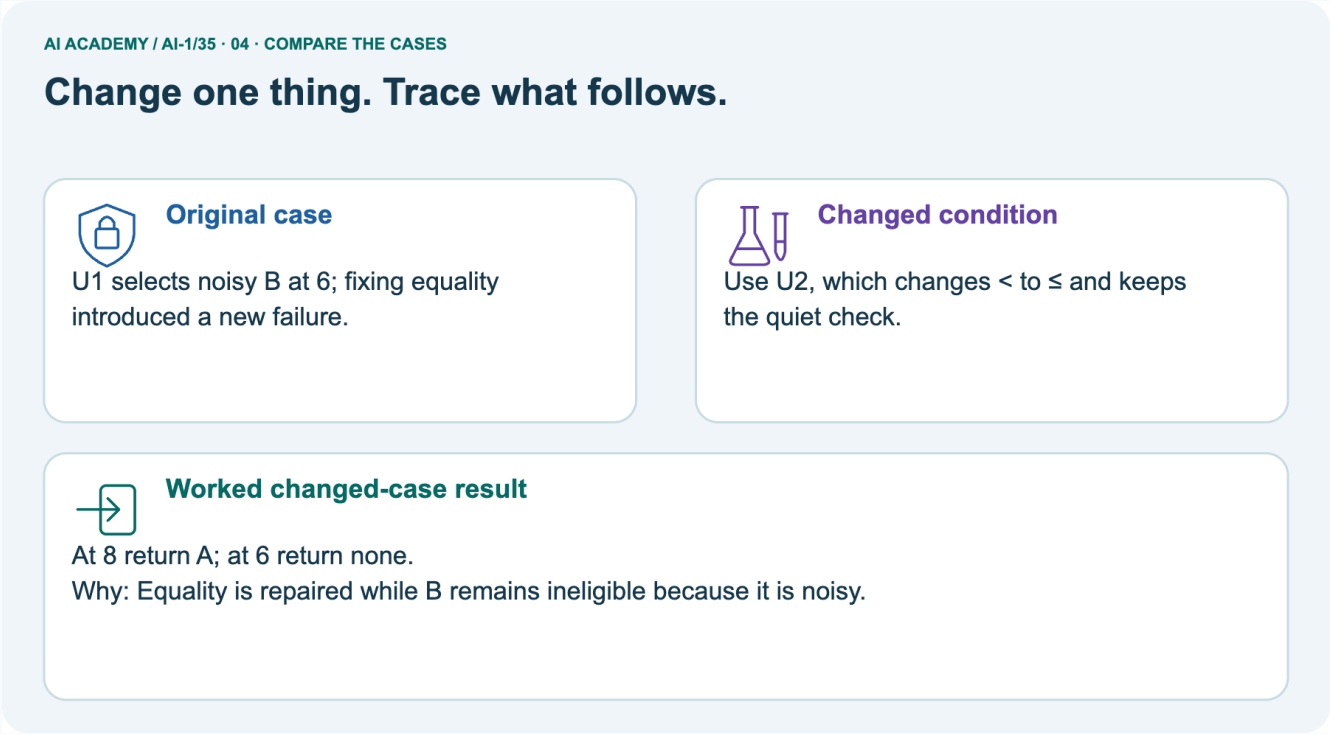
A **repair** with **evidence**. Keep the failing trace | Save a new version | Rerun the revealing case. Name the broken requirement | Correct the faulty step | Rerun earlier passing cases.

Keep **input** and catalogue labels | Record what changed | Keep every result and limit

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

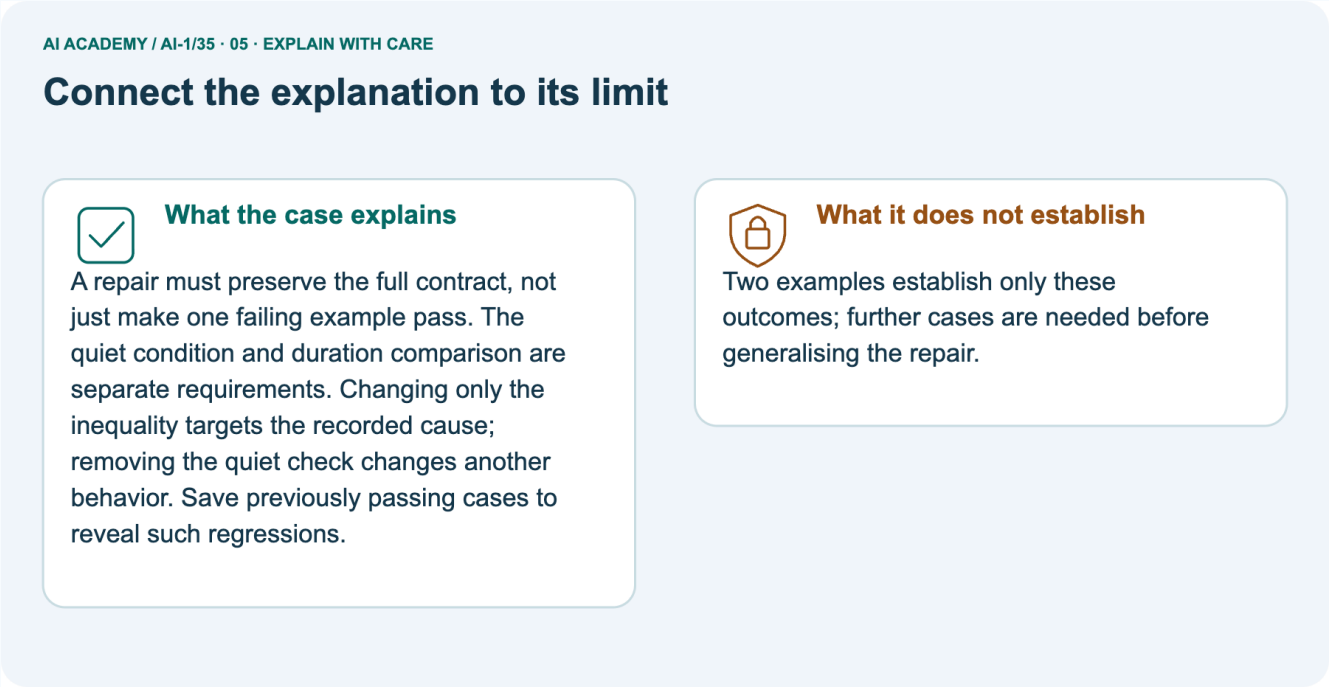
Original case: U1 selects noisy B at 6; fixing equality introduced a new failure. Changed condition: Use U2, which changes < to ≤ and keeps the quiet check. Expected result: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

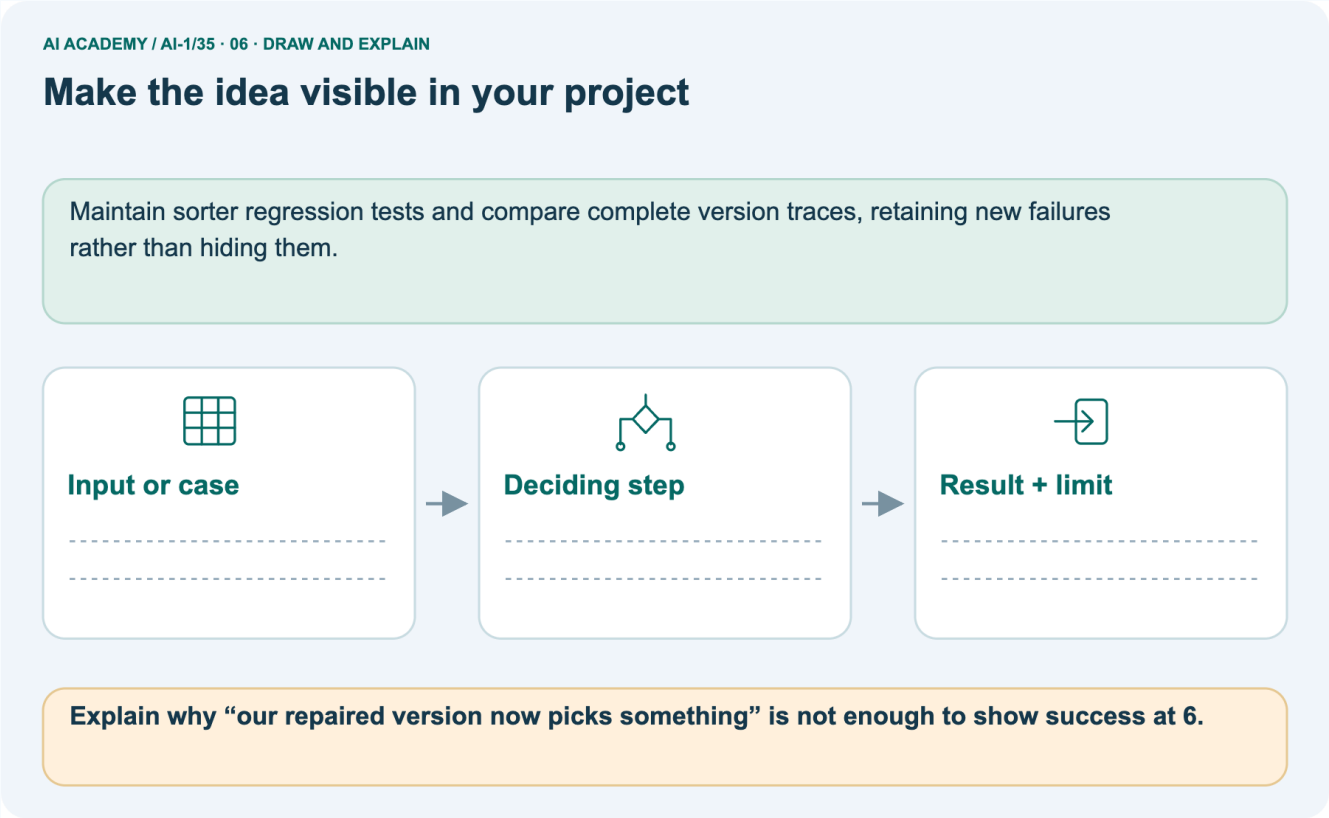
Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

Limit: Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them. Draw your **input**, deciding step, result and limit.

Supported question: Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Improve One Cause: the case

02. Follow the deciding steps

03. A repair with evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.